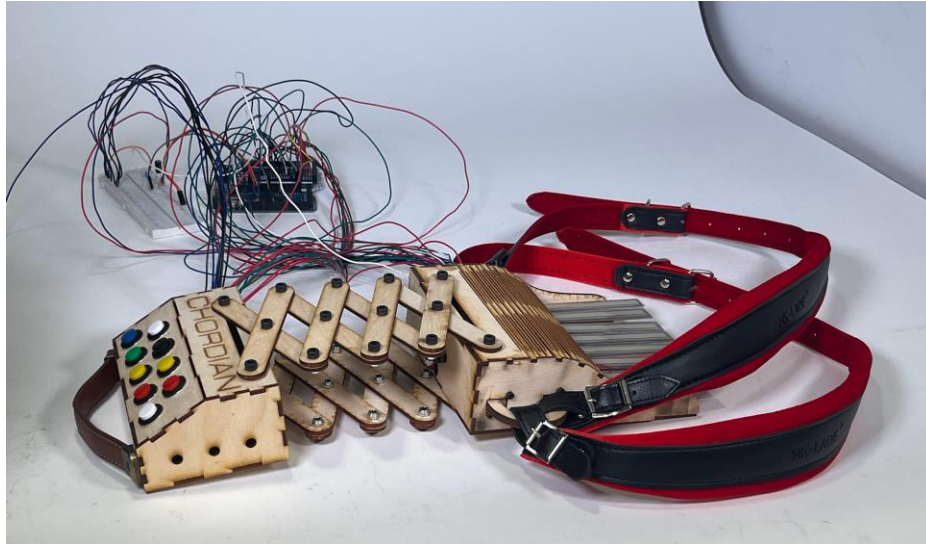


The CHORDIAN



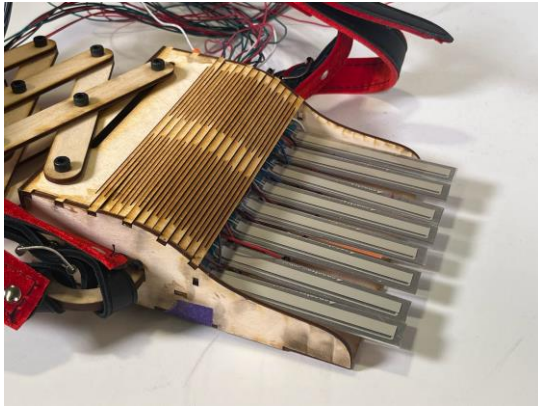
Summary

The Chordian is an electronic “evolution” of an accordion. Just like a normal accordion, the player has access to multiple forms of unique expression. Our group strived to create an instrument that could be performed by itself, with multiple options for chords and singular notes. Additionally, the Chordian uses multiple types of sensors, each with different parameters to control. Overall, the Chordian successfully pairs the user’s multi-dimensional input with dynamic and musical generation.

Instrument Layout and Playing

The Chordian is broken down into three main parts: the key box, the button box, and the linkage. The key and button boxes are connected by the linkage system to create a fully functioning accordion motion. Additionally, there are accordion straps connected to the key box to allow for further stability and playing ease.

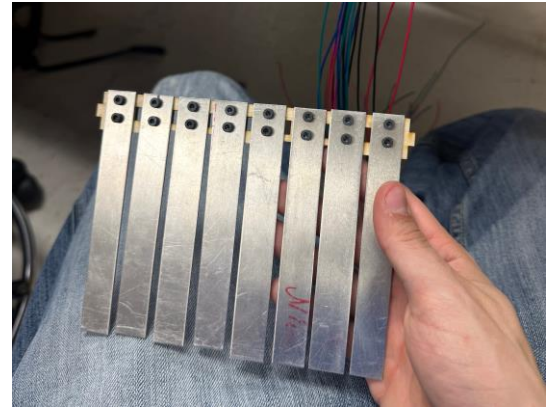
Materials and Construction



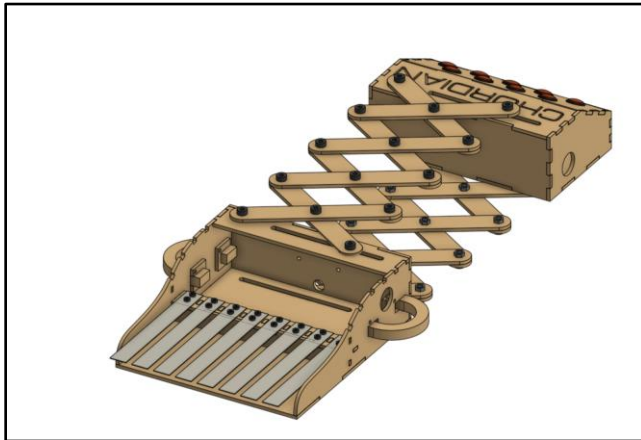
The key box is composed of 8 metal keys, made out of aluminum sheet metal. Each key was cut and sized using the jump shear and the drill press was used to create the holes. On each key is one linear soft potentiometer and one bend sensor. On the outside of the box, there is a distance sensor attached. The sensors are soldered onto the protoboard that lives within the key box. The actual box is made out of laser cut wood, 3mm thick.

The linkage attaching the two boxes is also made out of laser cut birch wood, but also consists of multiple bolts, nuts, and washers to ensure smoothness when moving the boxes.

The button box, also made out of laser cut wood, consists of 9 buttons. Attached is a leather strap to provide stability and increase ergonomics of the instrument.

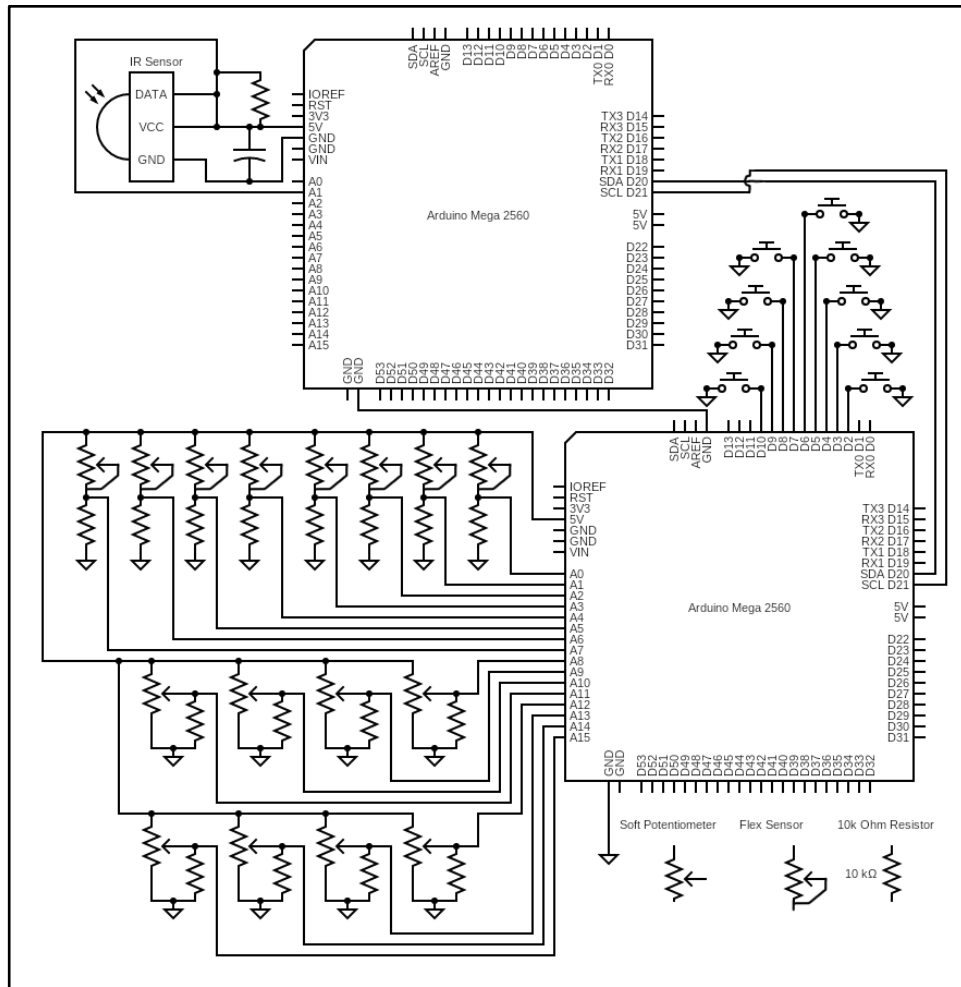


All of the laser cut pieces were first designed in OnShape, where we took advantage of designing with variables that we could easily change to quickly update our design choices.



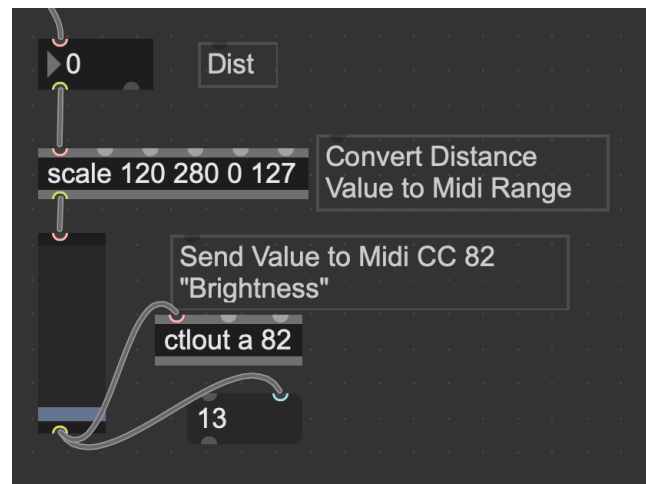
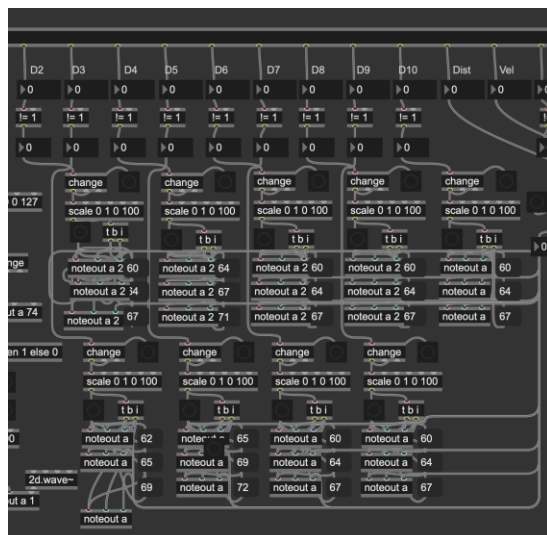
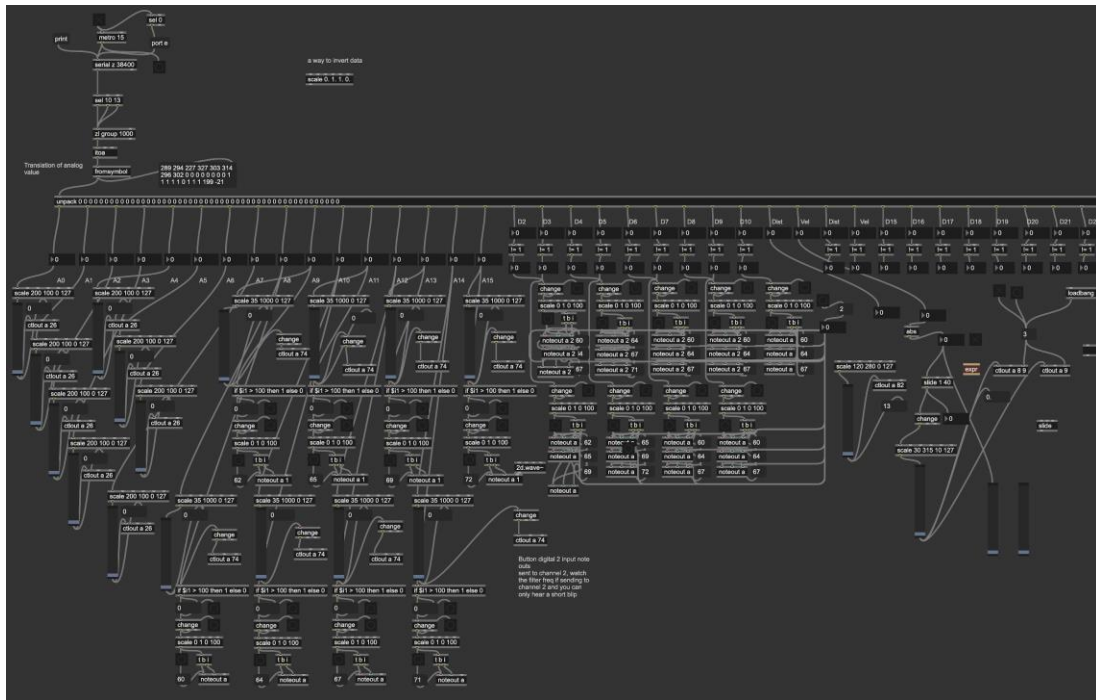
Sensors and Circuitry

To read our 8 soft potentiometers, 8 bend sensors, 1 distance sensor, and 9 buttons, we needed to hook them up to an Arduino (Mega). For the soft potentiometers, there are three pins: one receiving 5 volts, one into ground, and the last one being sent to an analog pin and also being pulled down to ground by a 10 K ohm resistor. By wiring it this way, we create a voltage divider and decrease the noise the potentiometer produces. The bend sensors are also very similar, although only having two pins: one receiving 5 volts and the other going into an analog pin and also a pulldown resistor. The distance sensor also uses an analog pin, thus the need for another Arduino (one Mega has 16 analog pins). By using the SDA and SCL pins, we can connect our two Arduinos together. The distance sensor has three pins, receiving 5 volts, ground, and an analog pin. We also added a resistor and capacitor to deal with noise and to smooth out the values. Lastly, the buttons do not act as a resistor, so we used arduino's built in pull up resistors within the digital pins. Each button goes to a different digital pin and to ground. The wiring diagram is pictured below:



To better organize and solidify our wiring, we soldered all of the sensors to the protoboard, except for the wires going into the arduino (analog pins, digital pins, 5V, GND). By having longer wires, we could more easily move our linkage system without worrying about tangling or detaching any wires.

Lastly, the distance sensor. There are currently two different ways we are considering to convert the values into expressive control. One way is to use the distance between the two boxes to control the amplitude. Another way is to use the acceleration and velocity to control the amplitude. As you accelerate the boxes away or towards each other, the volume swells. The entire Max patch is pictured below:



The Reason patch used for the Chordian is a simple one. Using two different NN19 modules, we send the soft potentiometer's note on messages to channel 1 and the button presses to channel 2. This way, the chords and singular notes can be played together and controlled individually.

Problems / Future Notes

The first main problem to address is the bend sensors. Although they are working perfectly and sending signals, they do not provide enough of a range of values to effectively use them. To allow for more bending, I would build an additional fulcrum for each key to emphasize the bending of the sensor, allowing for more convincing values. Secondly, instead of bend sensors, I would also consider changing them out for FSR's or another sensor. I think the bend sensors limit our decisions when creating this instrument, so it would be interesting to see another sensor.

Another main issue we ran into was sending the key box and the button box to two separate midi channels. Although it seems to be simple on Max, sometimes notes on messages were sent to both channels, introducing some interference and odd sounding notes. In the future, I would spend more time with Max and Reason to ensure that chords and singular notes could be played at the same time and even have different patches/samples.

If we had more time with this project I would add several more features. First being rotary encoders for changing the scale. This would change the root note and thus change the notes being played on the keys and the chords being played. Similarly, another rotary encoder could change a parameter like LFO amount. We laser cut holes for the rotary encoders, but didn't have enough time to incorporate them. Another feature to add is 2 buttons to go up and down octaves. I would also like to add more buttons/pads to play samples and drone notes. It would also be interesting to put a looper/drum machine on the button box.

In the future, more time could be used to make the Reason patch more complicated and intricate. We did not work too much on sound design, so it would be good to come up with several Chordian specific patches that match the "vibe" of our instrument.

Finally, with more time, I would work on using WiFi or Bluetooth to decrease the amount of long wires sticking out of the instrument. By redesigning the housing (possibly a

metal housing), we could stick the entire Arduino in the key/button box and send via Bluetooth. This would enable the player much more expressive freedom when performing.